

Resonant versus geometric anisotropy of the Purcell enhancement near a gold nanorod: plasmon-mode decomposition, an equal-volume control and design rules for single-photon sources

Amin Hossein Sadidi¹, Seyed Mohammad Parsa Molaei Tabari¹, and Malek Bagheri Harouni^{*1}

¹Department of Physics, Faculty of Science, University of Isfahan, Isfahan, Iran

Abstract

Near the tip of a gold nanorod the decay rate of a quantum emitter depends strongly on the orientation of its transition dipole. Part of this anisotropy is generic—any curved metal surface quenches a perpendicular dipole more strongly than a parallel one—and part is caused by the longitudinal plasmon resonance. We separate the two for a capped gold nanorod (60 nm × 20 nm) with boundary-element calculations that match the exact Mie solution for the equal-volume sphere to 1 %, and with a plasmon-eigenmode decomposition. At a 5 nm gap the axial dipole reaches a Purcell factor of 1.6×10^3 and a radiative enhancement of 117, and about 90 % of the enhancement is carried by the longitudinal dipolar mode. The non-resonant part has an axial-to-transverse ratio of about three, like the equal-volume sphere, and is set by the tip curvature (2 for flat ends, 8 for sharp tips); the resonance raises the ratio to 49 for the total and 329 for the radiative rate, with a maximum near 10 nm that grows from 19 to 872 between aspect ratios 2 and 5. We derive design rules for single-photon sources—the optimal gap grows with the emitter’s intrinsic quantum yield, the rod should lie in the focal plane—and show robustness to emitter position, substrate, nonlocal response and tip shape. Single emitters remain weakly coupled (at most 37 % of the strong-coupling threshold), so single-photon statistics are preserved, and at cryogenic temperatures the antenna raises photon indistinguishability above 96 %. Without adjustable parameters, the model reproduces the thousandfold single-molecule fluorescence enhancement measured near gold nanorods. Finite-difference time-domain solvers reproduce the radiative quantities but not the total rate in the nanometre gap.

Keywords: Purcell effect; local density of optical states; gold nanorod; plasmon eigenmodes; quenching; single-photon source; boundary element method.

1 Introduction

The spontaneous emission rate of a quantum emitter is not an intrinsic property but depends on its electromagnetic environment [1, 2]. Metallic nanoparticles supporting localized surface plasmon resonances confine light to volumes far below the diffraction limit and can enhance emission rates by several orders of magnitude [3–5]. This has made plasmonic nanoantennas a candidate platform for bright, fast single-photon sources [6–9], in which a large rate enhancement shortens the excited-state lifetime and raises the achievable repetition rate, and a short radiative lifetime relative to the dephasing time is a prerequisite for indistinguishable photons [10].

^{*}Corresponding author: [\[e-mail\]](#)

The price of plasmonic confinement is absorption. At emitter–metal separations of a few nanometres, energy is transferred to non-radiative channels in the metal, and the fluorescence is quenched rather than enhanced [11–14]. The total decay rate and the radiative rate therefore have to be treated separately, and their ratio—the antenna quantum efficiency—depends sensitively on distance, wavelength and the intrinsic quantum yield of the emitter [15–17].

The optical response of metal nanoparticles is governed by their size, shape and dielectric environment, which set the frequency, width and near-field distribution of the localized resonances [3, 18]. For emitters within a few nanometres of the metal, two further regimes become relevant and are the subject of quantum plasmonics [19]: when the emitter–plasmon coupling rate exceeds the plasmon and emitter losses, the system enters strong coupling, which has been reached for single molecules only in sub-nanometre gaps of particle-on-mirror cavities [20, 21]; and below about 1 nm electron spill-out, nonlocality and tunnelling modify the metal response itself [22–24]. Which of these regimes applies sets the appropriate description of the emitted light: a weakly coupled emitter keeps its quantum statistics and is described by modified decay rates, whereas strong coupling produces hybrid light–matter states.

Gold nanorods are attractive antennas because their longitudinal plasmon resonance is tunable through the aspect ratio across the visible and near infrared [25, 26], they can be synthesised with good control [27], and their tips concentrate the near field. Single-molecule experiments have demonstrated fluorescence enhancements exceeding a thousandfold near individual gold nanorods [28, 29], down to emitters with quantum yields of 10^{-4} [17]. On the theory side, Mohammadi *et al.* computed radiative enhancements and antenna efficiencies for gold nanorods and nanospheroids as a function of aspect ratio, volume and embedding medium [30, 31], and body-of-revolution solvers have recently been extended to excitation and emission near core–shell rods [32]. These studies, which exploit the rotational symmetry of an emitter on the rod axis, consider the axial dipole and gaps of 10 nm or more.

A consequence of the nanorod’s elongated shape is a strong dependence of the response on the orientation of the emitter’s transition dipole. Yet a dipole close to any metal surface—a sphere or a plane—also shows an orientation dependence: in the quasi-static near field, energy transfer to the metal is twice as strong for a dipole perpendicular to the surface as for a parallel one [13]. It is therefore not obvious how much of the anisotropy near a nanorod tip is generic and how much is caused by the longitudinal resonance that distinguishes the nanorod from a compact particle. The distinction matters in practice: emitters such as colloidal quantum dots, nanodiamonds or dye molecules are often randomly oriented, and orientation-averaged or scalar descriptions—a single “Purcell factor”—discard precisely the resonant part. A second, methodological issue is that the total rate in a nanometre gap is difficult to converge with volume-discretisation methods, whereas the radiative rate is not; conclusions about quenching then depend on the solver.

In this work we separate the two contributions in two independent ways. First, we compare the nanorod with an equal-volume gold sphere, probed with axial (radial) and transverse (tangential) dipoles at the same gap. Second, we expand the decay rate in the quasistatic plasmon eigenmodes of the particle [33] and isolate the contribution of the longitudinal dipolar mode from the non-resonant background. All rates are computed with a boundary-element method validated against the exact Mie solution, and checked against two finite-difference time-domain solvers. We then translate the results into design rules for plasmon-enhanced single-photon sources: the dependence of the optimal gap on the emitter’s intrinsic quantum yield, the tolerance to its position, the effect of a substrate and of nonlocal response, and resonance-matched antennas for representative emitters.

2 Theory

2.1 Decay rates and efficiency

For a point dipole $\mathbf{p} = p \hat{\mathbf{n}}$ at $\mathbf{r}_0$ in the weak-coupling regime, the total decay rate normalised to its value in the homogeneous host is [2]

$$\frac{\Gamma_{\text{tot}}}{\Gamma_0} = \frac{6\pi c}{\omega} \hat{\mathbf{n}} \cdot \text{Im} \overleftrightarrow{\mathbf{G}}(\mathbf{r}_0, \mathbf{r}_0; \omega) \cdot \hat{\mathbf{n}} \equiv F_p(\omega), \quad (1)$$

where $\overleftrightarrow{\mathbf{G}}$ is the dyadic Green function. Classically, $F_p = P_{\text{tot}}/P_0$ is the power delivered by a constant-amplitude dipole relative to the homogeneous host. The power radiated to the far field, $T = P_{\text{rad}}/P_0$, gives the radiative enhancement, and $F_p - T$ is absorbed in the metal. For an emitter of intrinsic quantum yield q_0 , the quantum efficiency in the presence of the antenna is [15]

$$\eta(q_0) = \frac{T}{F_p + (1 - q_0)/q_0}, \quad (2)$$

which reduces to the antenna efficiency $\eta_a = T/F_p$ for $q_0 = 1$. The total decay rate of the emitter is multiplied by $q_0 F_p + 1 - q_0$.

For a dipole on the symmetry axis of a body of revolution the Green tensor is diagonal, with an axial component and two equal transverse ones; the anisotropy is fully characterised by the ratios $F_p^{\parallel}/F_p^{\perp}$ and $T^{\parallel}/T^{\perp}$. For a sphere, axial and transverse correspond to radial and tangential dipoles, for which exact Mie expressions exist [34–36]. The equal-volume sphere therefore provides a reference that has the same amount of gold and the same dipole–surface gap, but no elongated resonance.

2.2 Plasmon-eigenmode decomposition

In the quasistatic limit the surface charge σ induced on a particle obeys the boundary integral equation $(\Lambda + F)\sigma = -\phi'_{\text{ext}}$, where F is the normal derivative of the Coulomb kernel on the surface, ϕ'_{ext} the normal derivative of the external (dipole) potential, and $\Lambda(\omega) = 2\pi(\varepsilon + \varepsilon_b)/(\varepsilon - \varepsilon_b)$ contains the permittivities of the metal and the host [33, 37]. Expanding in the right and left eigenvectors of F , $Fu_k = \lambda_k u_k$, $\tilde{u}_k F = \lambda_k \tilde{u}_k$, $\tilde{u}_k u_l = \delta_{kl}$, gives

$$\sigma = - \sum_k \frac{u_k (\tilde{u}_k \cdot \phi'_{\text{ext}})}{\Lambda(\omega) + \lambda_k}, \quad F_p - 1 = \sum_k \Delta F_p^{(k)}, \quad (3)$$

because the induced field at the dipole, and hence $F_p - 1$, is linear in σ . Each mode resonates where $\text{Re} \Lambda(\omega) = -\lambda_k$, and its weight is the product of its coupling to the dipole ($\tilde{u}_k \cdot \phi'_{\text{ext}}$) and of the field it produces back at the dipole. For the rod, the longitudinal dipolar mode is identified as the eigenvector with the largest axial oscillator strength. On the axis, it couples only to the axial dipole, whereas the transverse dipole couples only to modes with azimuthal number $m = \pm 1$. We call $\Delta F_p^{(L)}$ the resonant part and $F_p - 1 - \Delta F_p^{(L)}$ the background; the background contains the high-order modes that accumulate near the surface-plasmon condition and describe the lossy-surface (quenching) channel [11, 13].

3 Methods

3.1 Geometry and material

The nanorod is a gold cylinder of length 40 nm and radius $R = 10$ nm terminated by two hemispherical caps (total length $L = 60$ nm, aspect ratio 3), a geometry accessible by seed-mediated growth [27]. The dipole is placed on the rod axis at a distance d from the apex ($d = 5$ nm unless stated otherwise; 3–30 nm in the gap study). The equal-volume sphere has $R_{\text{sph}} = 15.874$ nm.

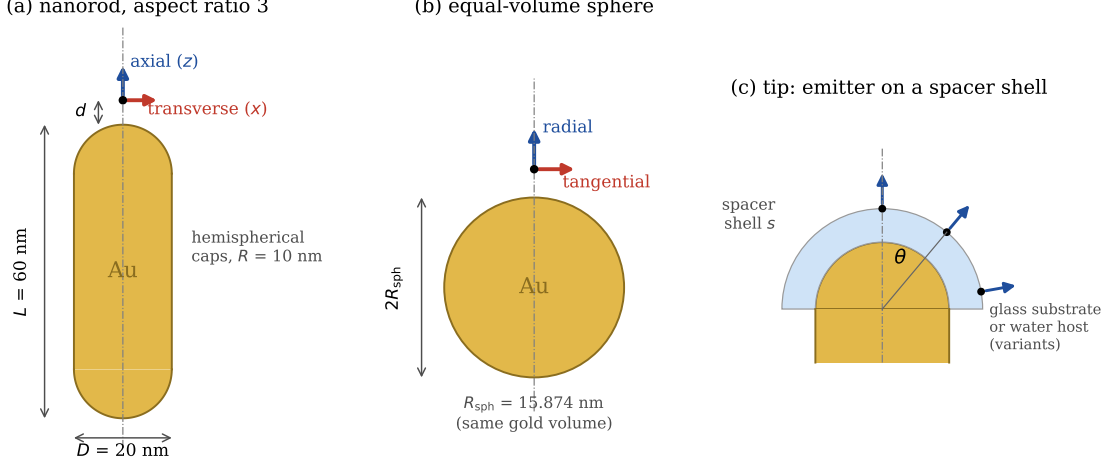


Figure 1: Geometry. (a) Capped gold nanorod ($L = 60$ nm, $D = 20$ nm) with an emitter on the axis at a gap d from the apex, oriented along (axial, z) or perpendicular to (transverse, x) the rod axis. (b) Gold sphere of the same volume, probed with radial and tangential dipoles at the same gap. (c) Emitter on a spacer shell of thickness s around the tip, at polar angle θ , with its dipole normal to the surface; variants with a glass substrate and in water are also studied.

The host is air unless stated; Fig. 1 summarises the geometry and the dipole orientations. Gold is described by the tabulated Johnson–Christy permittivity [38] in all boundary-element and Mie calculations.

3.2 Boundary-element method

The rates are computed with the boundary-element method (BEM) of the MNPBEM17 toolbox [37, 39, 40], which solves the full Maxwell equations for the surface charges and currents and discretises only the particle boundary, with curved boundary elements. F_p follows from the induced field at the dipole and T from the far field, both in units of the rate in the homogeneous host. The rod surface is discretised with element sizes $h = 3, 2$ and 1.5 nm (720, 1550 and 2829 elements), the sphere with 400, 784 and 1444 vertices; all gaps of one geometry are solved simultaneously. The eigenmode decomposition, Eq. (3), uses the 40 lowest quasistatic eigenmodes of the 2 nm mesh; the full quasistatic F_p is computed with the same mesh. Additional calculations, described where they are used, include a glass substrate (layer Green function [39]), the hydrodynamic nonlocal response through the equivalent local cover layer of Luo *et al.* [41], plane-wave excitation, other shapes and silver. All calculations ran on GitHub-hosted runners; the scripts are provided.

3.3 Finite-difference time-domain cross-checks

The rod and sphere were also computed with a commercial three-dimensional FDTD solver (Lumerical FDTD 2024 R1, 2 nm override mesh) [42, 43] and with a body-of-revolution FDTD in cylindrical coordinates (Meep [44], grids of 2, 1.33 and 1 nm), in which an axial dipole couples only to the $m = 0$ field harmonic. Details, the gold Drude–Lorentz fit required by the time-domain solvers and all benchmarks are given in the Supplementary Information.

4 Results

4.1 Validation and convergence

For the equal-volume sphere at a 5 nm gap the BEM reproduces the exact Mie solution for both dipole orientations over 500–900 nm (Fig. S3): with 1444 vertices the median deviation is 0.9 % for the radial F_p and 1.4 % for the radial T , 0.7 % and 1.0 % for the tangential dipole, and at most 4 %; the deviation is of the same size at 10 and 20 nm and in water (Table S1). For the rod (axial dipole, 5 nm gap) the peak values change by 0.3 % (F_p) and 0.4 % (T) between the 2- and 1.5 nm meshes, and the peak positions not at all; a 1 nm mesh (6324 elements, 500–808 nm) changes them by a further 0.2 % (Table S2, Fig. S1). We quote the 1.5 nm mesh. The three-dimensional FDTD reproduces the radiative peak (119 at 616 nm, BEM 117 at 608 nm) but gives a 9 % lower F_p (1496 versus 1638); the body-of-revolution FDTD reproduces T to 7 % on a 1 nm grid but does not converge for F_p in the 5 nm gap, where the staircased surface overestimates it by about a factor of two even for the sphere (Supplementary Information). Volume discretisation is therefore adequate for the radiated power and the emission pattern, but the total rate—and hence the quenching—in a nanometre gap requires a method that represents the metal surface exactly.

4.2 Axial versus transverse dipole, rod versus sphere

Figure 2 shows F_p , T and η_a at a 5 nm gap. For the axial dipole at the rod tip F_p peaks at 1638 at 604 nm and T at 117 at 608 nm, where $\eta_a = 7.3\%$: more than 92 % of the energy is absorbed. A second F_p maximum at 512 nm (848) coincides with the onset of interband absorption and is almost purely non-radiative ($T = 2.0$). The transverse dipole at the same position has $T < 0.5$ at all wavelengths—it radiates less than in free space—and its F_p maximum (248 at 508 nm) is non-radiative.

The equal-volume sphere separates the generic from the resonant contribution (Fig. 3). Its radial-to-tangential ratio of F_p lies between 2.75 and 2.97 over the whole band, in agreement with the exact solution (2.76–2.97) and near the quasi-static flat-surface value of two, without resonant structure. The rod shows a similar ratio (3.3–4.0) below 530 nm but rises to 49 at the longitudinal resonance. For the radiated power the sphere ratio is 11–53, varying slowly with wavelength, because a tangential dipole near a metal surface radiates poorly; the rod ratio is 5–6 below 530 nm and 329 at resonance. At 608 nm the rod exceeds the sphere by factors of 15 in F_p and 23 in T , whereas the maxima of the two particles differ by only 2.0 in F_p : with the same amount of gold, the elongated particle does not provide a larger total rate but moves a radiative resonance out of the interband region.

4.3 Mode decomposition

The eigenmode analysis makes this separation explicit (Fig. 4). The lowest non-trivial mode of the rod is the longitudinal dipole ($\lambda_k = -5.01$, quasistatic resonance at 595 nm, 13 nm to the blue of the retarded result); the transverse dipolar modes lie at $\lambda_k \approx -1.71$, which requires $\varepsilon \approx -1.75$ and therefore falls into the strongly damped interband region below 500 nm. At a 5 nm gap and the resonance, 91 % of $F_p - 1$ is carried by the longitudinal mode alone. The remaining background is a smooth function of wavelength and its axial value exceeds the transverse rate by a factor of 3.1, close to the sphere’s ratio: *the geometric anisotropy of the rod is that of any curved metal surface, and the excess is the contribution of a single resonant mode.*

The two contributions depend differently on the gap (Fig. 4b,c). The background decays as the near-field of a dipole above a lossy surface, approximately as d^{-3} , whereas the coupling to the dipolar mode decays on the length scale of the tip radius. The share of the dipolar mode therefore increases from 80 % at 3 nm to 96 % at 10 nm and 98 % at 20 nm, and the total anisotropy $F_p^{\parallel}/F_p^{\perp}$ is non-monotonic: in the full retarded calculation it is 19 at 3 nm, 49 at 5 nm, 136 at 10 nm, 128 at 15 nm and 75 at 20 nm. At small gaps the quenching background, which both orientations share,

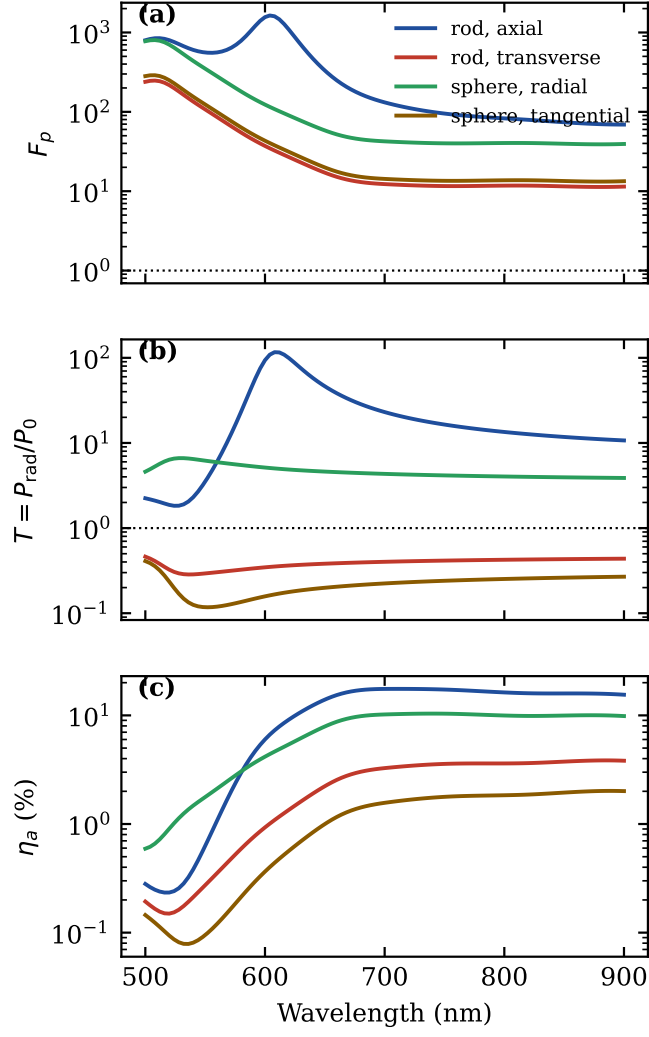


Figure 2: (a) Purcell factor, (b) radiative enhancement and (c) antenna efficiency at a 5 nm gap for the rod with axial and transverse dipole and the equal-volume sphere with radial and tangential dipole (BEM). Dotted: free space.

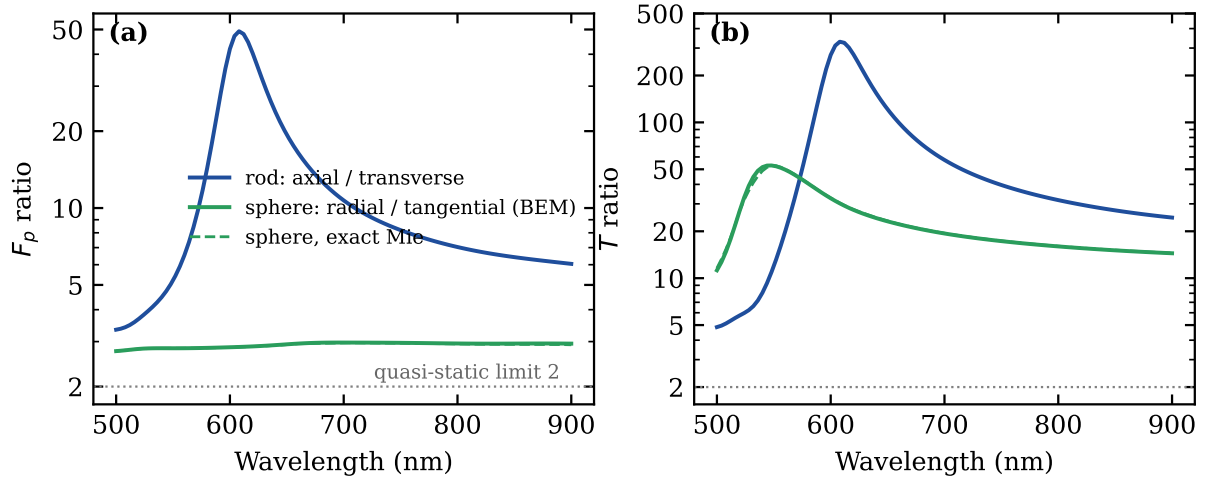


Figure 3: Orientation ratios of (a) F_p and (b) T for the rod (axial/transverse) and the equal-volume sphere (radial/tangential) at a 5 nm gap; dashed: exact Mie solution for the sphere. Dotted: quasi-static flat-surface limit of 2.

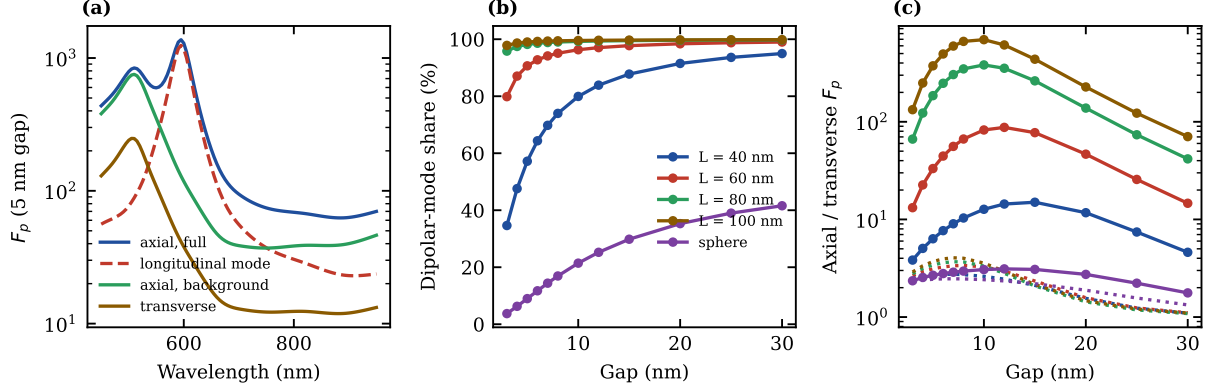


Figure 4: Plasmon-eigenmode decomposition (quasistatic BEM). (a) F_p of the axial dipole at a 5 nm gap, the contribution of the longitudinal dipolar mode, the remaining background and the transverse F_p . (b) Share of $F_p - 1$ carried by the longitudinal dipolar mode at its resonance versus gap. (c) Axial/transverse ratio of F_p (solid) and of the background alone (dotted) versus gap.

dilutes the resonant contribution; at large gaps the transverse rate approaches its free-space value while the coupling to the mode keeps decreasing. The same decomposition for aspect ratios 2–5 and for the sphere (Fig. 4b, Table S3) shows that the background ratio is a property of the curved metal surface rather than of the shape: it is 2.5–3.7 for all four rods and 2.3–2.4 for the sphere at gaps of 3–10 nm. What changes with the shape is the weight of the dipolar mode. For the sphere its resonance lies in the interband region and it carries only 9 % of $F_p - 1$ at 5 nm; for the rods its share at 5 nm rises from 57 % (aspect ratio 2) to 91 %, 98 % and 99 % (aspect ratios 3, 4, 5), as the resonance moves away from the interband edge and gains oscillator strength. In the retarded calculation the maximum anisotropy, always found near 10 nm, accordingly grows from 19 (aspect ratio 2) to 54, 136, 621 and 872 (aspect ratios 2.5, 3, 4, 5).

The decomposition is quasistatic, but its main conclusion carries over to the retarded problem. Fitting the full BEM spectra with a single complex pole, the quasinormal-mode form of a modal contribution [45, 46], plus a smooth background gives the share of the dipolar mode at its resonance directly: 88 % for aspect ratio 3, 97 % for aspect ratio 4 and essentially 100 % for aspect ratio 5 at a 5 nm gap, with linewidths of 133, 87 and 85 meV (Table S4). Applied to the quasistatic spectra, where the exact answer is known, the same fit reproduces the mode share to within 1–2 percentage points for aspect ratios 4 and 5 and underestimates it by about 13 points for aspect ratio 3, whose resonance is closer to the interband edge; for aspect ratio 2 the fit cannot separate mode and background. The background itself is the small difference of two large numbers and is therefore taken from the quasistatic analysis.

4.4 Gap dependence and the quantum-yield design rule

Increasing the gap from 3 to 20 nm (Fig. 5) leaves the radiative peak at 608–612 nm while F_p falls from 3782 (at the interband maximum, 508 nm) to 78 and T from 214 to 8.2. The antenna efficiency at the radiative peak increases only from 6.6 % to 11.6 %, whereas its spectral maximum, reached far to the red of the resonance, rises from 12 % to 60 %. The exact solution for the sphere shows the same trade-off (Fig. 5, lower row).

For a single-photon source the relevant figure of merit depends on the operating regime. Under saturated (continuous-wave or high-repetition) excitation the photon rate scales with the radiative rate, i.e. with T , for any q_0 . Under pulsed excitation with a period much longer than the excited-state lifetime the number of photons per pulse scales with $\eta(q_0)$, Eq. (2) [6, 7]. Figure 6 and Table 1 show that the optimal gap depends on q_0 . For $q_0 = 1$ the 20 nm gap is best (11.6 %) and the 3 nm gap worst (6.6 %). For $q_0 = 0.01$ the order is reversed beyond 7 nm: gaps of 7

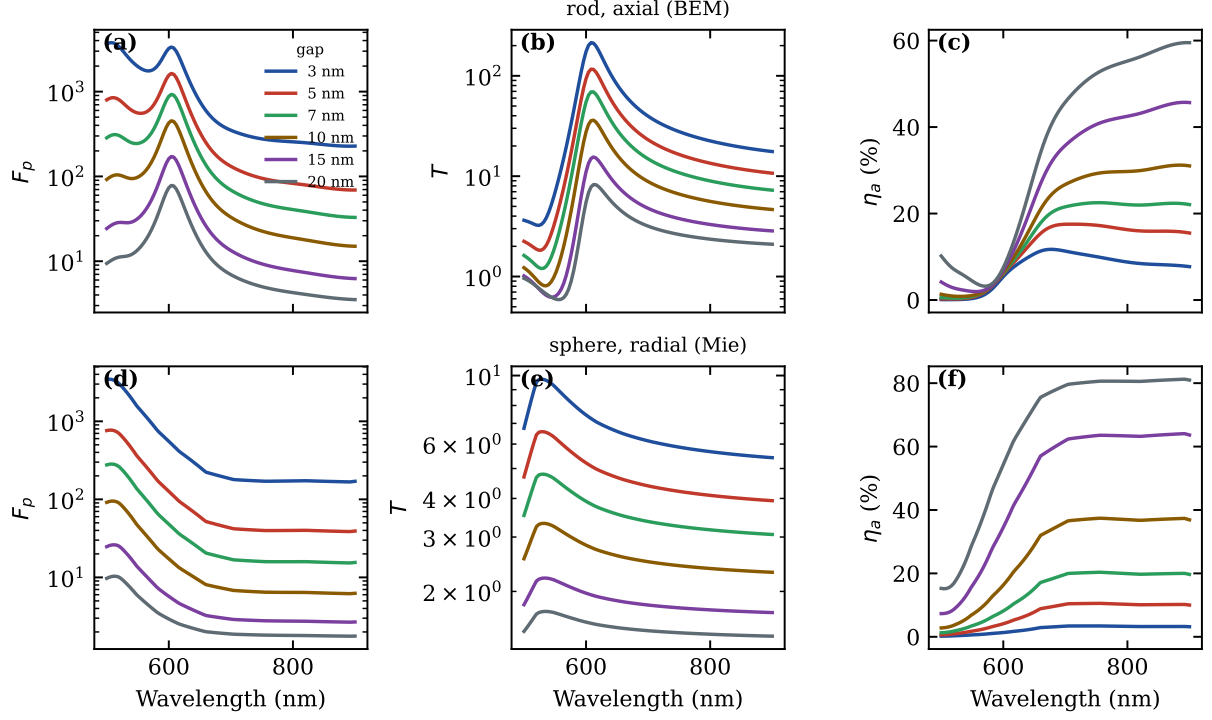


Figure 5: Gap dependence of F_p , T and η_a for the rod with axial dipole (top, BEM) and the equal-volume sphere with radial dipole (bottom, exact Mie).

Table 1: Quantum efficiency $\eta(q_0)$ at the radiative peak of the rod (axial dipole, BEM), Eq. (2).

gap (nm)	λ_T (nm)	$F_p(\lambda_T)$	$T_{\max}$	$q_0 = 0.01$	0.1	0.5	1
3	608	3265	214.1	0.064	0.065	0.066	0.066
5	608	1600	116.5	0.069	0.072	0.073	0.073
7	612	831	69.2	0.074	0.082	0.083	0.083
10	612	407	36.1	0.071	0.087	0.089	0.089
15	612	155	15.5	0.061	0.095	0.100	0.100
20	612	71	8.2	0.048	0.103	0.114	0.116

and 10 nm raise the efficiency from 1 % to 7.4 % and 7.1 %, whereas 20 nm gives 4.8 %. A large F_p is needed to outcompete the intrinsic non-radiative rate $(1 - q_0)/q_0$, but beyond that point additional F_p only feeds absorption, so the optimal gap increases with q_0 .

4.5 Excitation

Fluorescence below saturation is also proportional to the excitation rate, i.e. to the local intensity $|\mathbf{E} \cdot \hat{\mathbf{n}}|^2/|\mathbf{E}_0|^2$ at the emitter. For plane-wave illumination polarised along the rod axis the axial intensity enhancement at a 5 nm gap peaks at 119 at 610 nm (Fig. S2), equal within 2 % to the radiative enhancement $T = 117$ at 608 nm, as expected from reciprocity for a dipolar antenna [2, 5]. At common pump wavelengths it is small: 3.2 at 405 nm, 3.0 at 450 nm and 1.8 at 532 nm, compared with 3.9, 3.9 and 6.7 for the sphere, whose resonance lies near 530 nm. Unless the emitter is pumped near the longitudinal resonance—for example through an upper level of a molecule or quasi-resonantly for a quantum dot—the rod accelerates and redirects emission but contributes little to excitation; for emitters pumped far above the gap the excitation enhancement is set by the interband response and is no larger than for a compact particle.

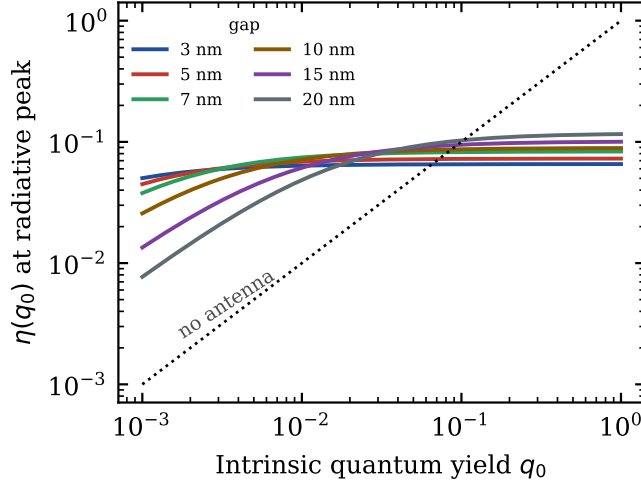


Figure 6: Quantum efficiency at the radiative peak versus intrinsic quantum yield, Eq. (2), for six gaps. Dotted: emitter without antenna ($\eta = q_0$).

4.6 Robustness: emitter position, substrate and nonlocal response

Emitter position. Emitters are often attached to a dielectric spacer shell of fixed thickness rather than placed exactly on the axis. We therefore moved the dipole over a shell of thickness s around the hemispherical cap, from the axis ($\theta = 0$) to the start of the cylindrical side ($\theta = 90^\circ$, a lateral offset of $R + s$), with the dipole normal (n) or tangential (t) to the surface (Fig. S4). For $s = 5$ nm the radiative enhancement of the normal dipole decreases only gradually, to 68 % of its on-axis value at $\theta = 90^\circ$ (70 % for $s = 10$ nm), and its efficiency from 7.6 % to 6.8 %, because the longitudinal mode carries charge over the whole cap. Tangential dipoles remain dark everywhere ($T \leq 1.2$). For a spacer-shell geometry the relevant tolerance is therefore the orientation of the transition dipole relative to the local surface normal, not the position on the cap.

Substrate. In experiments the rod usually lies on a glass cover slip. We placed the rod with its axis parallel to a glass half-space ($n = 1.52$) at a height of 1 nm (a ligand layer) and the dipole beyond the tip, on the rod axis and parallel to the interface. The longitudinal resonance shifts from 608 to 635 nm; at a 5 nm gap the radiated power rises to 266 and F_p to 2198 (both in units of the vacuum rate), and the antenna efficiency increases from 7.3 % to 12.1 % (14.3 % and 17.6 % at 10 and 20 nm). Relative to the same dipole on bare glass, whose rate is 1.27 times the vacuum rate, the enhancements are 209 (T) and 1.7×10^3 (F_p). The substrate therefore does not weaken the conclusions; it improves the efficiency, because the resonance moves further from the interband edge, and a dipole parallel to and close to a glass interface radiates predominantly into the glass [2], i.e. towards an oil-immersion objective (we did not resolve the angular distribution with the substrate). The dipoles perpendicular to the rod axis remain dark ($T < 2.3$).

Nonlocal response. The hydrodynamic correction, estimated with the equivalent cover layer of Luo *et al.* [41, 47, 48], blue-shifts the radiative peak by 5 nm and lowers F_p and T at resonance by 12–17 % at gaps of 5 and 10 nm; the antenna efficiency changes by less than 0.4 percentage points and the anisotropy increases slightly (from 49 to 70 at 5 nm), because the transverse, surface-dominated channel is suppressed more strongly. At 3 nm the non-radiative interband maximum of F_p is reduced by 27 %. These changes are of the order expected for the cover-layer model with the tabulated gold permittivity and do not affect the conclusions, but they indicate that absolute values of F_p at gaps of 3 nm and below carry an additional model uncertainty of this size.

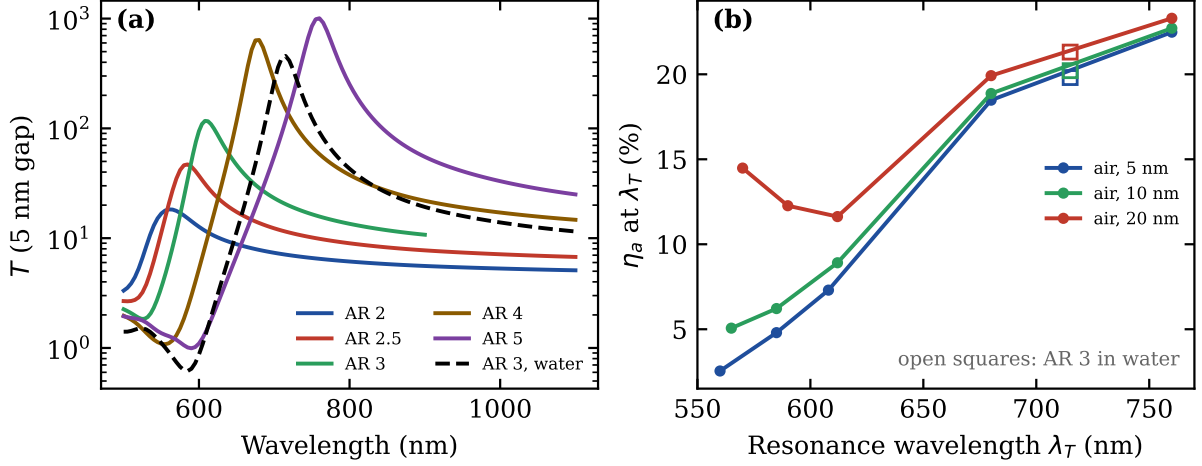


Figure 7: (a) Radiative enhancement of the axial dipole at a 5 nm gap for rods with $D = 20$ nm and aspect ratios (AR) 2–5 in air and AR 3 in water (BEM). (b) Antenna efficiency at the radiative peak versus resonance wavelength for three gaps; open squares: AR 3 in water.

4.7 Generality: aspect ratio, environment, size, shape and metal

Figure 7 shows the radiative enhancement and the antenna efficiency at the radiative peak for aspect ratios 2–5 ($D = 20$ nm) and for the aspect-ratio-3 rod in water. As the resonance moves from 560 nm to 760 nm, T at a 5 nm gap rises from 18 to 1.0×10^3 and η_a from 2.5 % to 22.5 %, because the resonance leaves the interband region; in water the same rod resonates at 715 nm with $T = 457$ and $\eta_a = 19.8$ %. These trends agree with those found by Mohammadi *et al.* at a 10 nm distance [30]; the efficiency at the radiative peak depends mainly on the resonance wavelength and only weakly on the gap between 5 and 20 nm. At fixed aspect ratio 3, the rod volume matters: for diameters of 15, 20 and 30 nm the efficiency at the radiative peak (5 nm gap) is 3.3 %, 7.3 % and 23.7 %, reflecting the growth of radiative damping with volume, while the resonance shifts only from 605 to 625 nm (14.0 % at 615 nm for $D = 25$ nm). A prolate spheroid with the same length and diameter resonates at 600 nm with $T = 106$ and $\eta_a = 5.3$ %; its sharper apex (radius of curvature 3.3 nm) raises the off-resonance ratio to 7.7, which shows that the geometric factor is set by the curvature of the surface on the scale of the gap. Real nanorods often have flatter ends than hemispheres. Replacing the hemispherical caps of the aspect-ratio-3 rod by half-spheroids with an axial semi-axis of 0.75, 0.5 and 0.35 times the radius (same length; Table S5) lowers the background ratio at a 5 nm gap from 3.1 to 2.6, 2.1 and 1.8, towards the flat-surface value of two, while the maximum anisotropy, 133–135 at 10 nm for all four shapes, is unchanged; the flatter ends red-shift the resonance by up to 15 nm and raise T from 118 to 157 and η_a from 7.6 % to 9.5 %. The geometric factor thus ranges from about 2 for flat ends to about 8 for sharp spheroidal tips, whereas the resonant factor is a robust property of the longitudinal mode. A silver rod of the same dimensions resonates at 505 nm with $T = 1.1 \times 10^3$ and $\eta_a = 38$ %, and its anisotropy reaches 677 in F_p . For all gold rods the non-resonant ratio, read at 500 nm, is 2.9–3.8 at a 5 nm gap, so the separation into a geometric factor of about three and a resonant factor that grows with the oscillator strength of the longitudinal mode holds generally.

4.8 Collection efficiency

The far field of the BEM solution, sampled on 1444 directions, gives the fraction of the radiated power that enters the cone of a collection objective. For the axial dipole at the radiative peak and a rod standing along the optical axis, this fraction is 1.2 % for $NA = 0.5$ and 18.1 % for $NA = 0.9$ (15.8–18.1 % for gaps of 5–20 nm), in agreement with the body-of-revolution FDTD

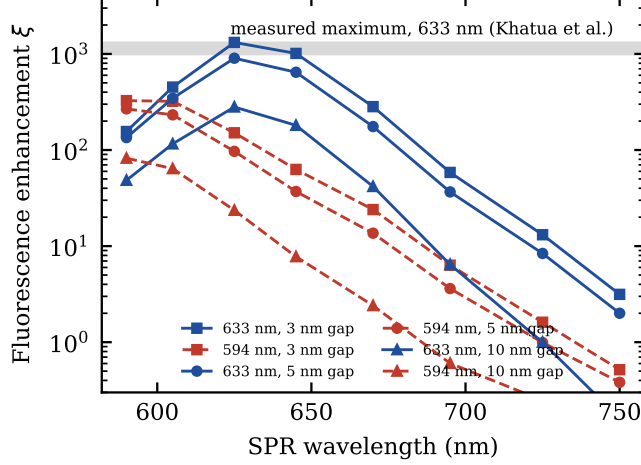


Figure 8: Fluorescence enhancement of crystal violet near gold nanorods of diameter 25 nm in glycerol (BEM, dipole on the axis and parallel to it) versus the computed resonance wavelength, for excitation at 633 and 594 nm and gaps of 3, 5 and 10 nm. Grey band: maximum enhancement measured by Khatua *et al.* [29] under 633 nm excitation.

(18.0 %) and within 7 % of a bare axial dipole (1.3 % and 19.4 %): the antenna multiplies the radiated power by two orders of magnitude but does not redirect it, in contrast to multi-element antennas designed for directivity [49]. If the same rod lies in the focal plane, the objective collects 9.4 % at NA = 0.5 and 32.5 % at NA = 0.9, the value of a dipole perpendicular to the optical axis (32.6 %). A rod standing along the optical axis therefore wastes almost half of its emission, and the orientation of the antenna relative to the collection optics is as important a design parameter as the gap. For a rod on glass, a dipole parallel to and close to the interface is expected to radiate preferentially into the substrate [2]; the angular split computed with the layer Green function was not numerically reliable for the weakly radiating transverse dipoles and is not reported.

4.9 Comparison with single-molecule experiments

The same method can be compared directly with the single-molecule experiments of Khatua *et al.* [29], who measured the fluorescence enhancement of crystal violet ($q_0 = 0.0228$, emission maximum 640 nm, lifetime 1.0 ns) diffusing in glycerol near individual gold nanorods of diameter 25 nm on glass, for longitudinal resonances between 560 and 720 nm and excitation at 594 and 633 nm. We modelled their rods (diameter 25 nm, lengths 39–70 nm) in a homogeneous medium of index 1.47, which represents glycerol on glass, with the dipole on the axis and parallel to it, as in their analysis. The fluorescence enhancement below saturation is $\xi = E_{\text{exc}}E_{\text{em}}$, with $E_{\text{exc}} = |E_z|^2/|E_0|^2$ at the laser wavelength and $E_{\text{em}} = \langle \eta(q_0) \rangle / q_0$ averaged over the emission band (Fig. 8). For the 47 nm rod, whose computed resonance (625 nm) is closest to the optimum, we obtain $E_{\text{exc}} = 127$ and $E_{\text{em}} = 7.1$ at a 5 nm gap, hence $\xi \approx 900$, and $\xi \approx 1.3 \times 10^3$ at 3 nm; Khatua *et al.* measured maximum enhancements of about 1100 for resonances near 630–650 nm and estimated $E_{\text{exc}} \approx 130$ and $E_{\text{em}} \approx 9$ with a discrete-dipole model. Our calculation also reproduces the spectral selectivity: under 633 nm excitation ξ falls by more than an order of magnitude for resonances at 695 nm and beyond, and under 594 nm excitation the maximum (270 at 5 nm) occurs for the shortest rods, compared with their estimate of 225. The computed total decay rate is shortened by factors of 11–27 for the best rods, consistent with the measured lifetime reduction by at least a factor of seven, which was limited by the instrument response. The agreement, obtained without adjustable parameters, supports the use of the same model for the design rules above. Because q_0 is small, this experiment probes the regime in which the antenna improves the efficiency itself; the near-unity-yield regime remains to be tested experimentally.

Table 2: Resonance-matched gold nanorods ($D = 20$ nm, air, axial dipole) for representative emitters. L : rod length maximising T at the emission wavelength λ_e ; speed-up: total decay rate relative to the bare emitter, $q_0 F_p + 1 - q_0$; η : Eq. (2).

emitter	λ_e (nm)	q_0	L (nm)	gap (nm)	T	speed-up	η	η/q_0
hBN defect [50]	580	0.87	50	5	46	950	4.2%	0.05
				20	3.9	39	8.8%	0.10
crystal violet [29]	640	0.02	60	5	58	10	11.2%	5.6
				20	5.7	1.4	7.8%	3.9
NV, 25 nm diamond [51]	690	0.10	80	5	436	222	19.6%	2.0
				20	30	14	21.7%	2.2
NV, 100 nm diamond [51]	690	0.70	80	5	436	1551	19.7%	0.28
CsPbI ₃ nanocrystal [52]	690	0.90	80	5	436	1993	19.7%	0.22
				20	30	116	23.2%	0.26

4.10 Design table for representative emitters

Table 2 combines the aspect-ratio sweep with literature values of the emission wavelength and the intrinsic quantum yield of four emitter classes: defects in hexagonal boron nitride ($q_0 = 0.87$ at 580 nm [50]), the dye crystal violet used in single-molecule nanorod experiments ($q_0 = 0.02$ [29]), NV centres in nanodiamonds, whose q_0 ranges from below 0.2 in 25 nm crystals to 0.1–0.9 in 100 nm crystals [51], and CsPbI₃ nanocrystals ($q_0 \approx 0.9$ [52]). For each emitter the rod length is chosen to maximise T at the emission wavelength. Three conclusions follow. First, the rate speed-up $q_0 F_p + 1 - q_0$ ranges from 10 to 2×10^3 and is largest for bright emitters, for which it is also accompanied by a loss of efficiency ($\eta/q_0 = 0.05$ – 0.33): for a near-unity emitter a gold nanorod is a device for speed and saturated brightness (through T), not for efficiency. Second, for poor emitters such as crystal violet the antenna improves the efficiency itself, by a factor of 5.6 at gaps of 5–10 nm. Third, the emission wavelength matters more than the gap: an emitter at 690 nm coupled to a rod resonating at 675 nm reaches $\eta_a \approx 20\%$ and $T > 400$, five times the radiative enhancement available at 580 nm, where interband absorption limits both.

4.11 Quantum regime: coupling strength, indistinguishability and photon statistics

The rate-based description used so far assumes weak coupling. For a single plasmon mode of energy width $\hbar\kappa$ in the bad-cavity limit, the mode-induced decay rate is $\Gamma_{\text{mode}} = 4g^2/\kappa$, so that the emitter–plasmon coupling follows from the modal Purcell factor as $\hbar g = \frac{1}{2}\sqrt{F_p^{\text{mode}} \hbar\Gamma_0 \hbar\kappa}$, and strong coupling requires $g > (\kappa + \gamma)/4$ [21], i.e. $F_p^{\text{mode}} > \kappa/(4\Gamma_0)$ for a spectrally narrow emitter. Taking F_p^{mode} and κ from the retarded pole fits and a radiative lifetime of 1 ns—an upper bound on Γ_0 for dyes, quantum dots and colour centres—we find $\hbar g = 5.3$ meV at a 5 nm gap for aspect ratio 3 ($\hbar\kappa = 133$ meV), 16 % of the threshold, and at most 37 % (7.9 meV, aspect ratio 5) for any rod and gap studied (Table 3). Strong coupling would require modal Purcell factors of 3×10^4 – 5×10^4 , one to two orders of magnitude above the present ones, consistent with the sub-nanometre gaps needed experimentally [20]; room-temperature emitter linewidths of tens of meV raise the threshold further. All emitters considered here are therefore weakly coupled, which justifies the rate description and implies that the antenna preserves the single-photon statistics of a single emitter. The main threat to antibunching is instead the photoluminescence of the gold itself, whose quantum yield is of order 10^{-6} [53] but which is excited by the same laser and has to be filtered spectrally or temporally.

The Purcell effect does change a quantum property of the emitted light, its indistinguishability. For an emitter with pure dephasing rate γ^* in the bad-cavity regime, $I = \Gamma/(\Gamma + 2\gamma^*)$ with

Table 3: Emitter–plasmon coupling for a radiative lifetime of 1 ns (axial dipole on the axis). F_p^{mode} and $\hbar\kappa$ from the retarded pole fits (Table S4); strong coupling requires $g > \kappa/4$.

aspect ratio	gap (nm)	F_p^{mode}	$\hbar\kappa$ (meV)	$\hbar g$ (meV)	$g/(\kappa/4)$
3	3	1908	145	6.7	0.19
3	5	1291	133	5.3	0.16
3	10	417	132	3.0	0.09
4	5	3481	86	7.0	0.33
5	5	4454	85	7.9	0.37

$\Gamma = F_p\Gamma_0$ ($q_0 = 1$) [54]. At room temperature ($\hbar\gamma^* \approx 10$ meV) I stays below 13 % even for $F_p = 4.5 \times 10^3$, so plasmonic nanorods cannot make dephased room-temperature emitters emit indistinguishable photons. At cryogenic temperatures ($\hbar\gamma^* = 1\text{--}10$ μeV), however, the same antenna raises I from 3–25 % for the bare emitter to above 96 % for $F_p \geq 80$ (20 nm gap) and above 98 % at the 5 nm gap, at the cost of the antenna efficiency of Fig. 6. For a cryogenic source of indistinguishable photons the trade-off is therefore between I , which favours small gaps, and the efficiency, which favours large ones.

5 Discussion

The orientation dependence of emission near a nanorod tip consists of two factors with different origins. A non-resonant factor describes how strongly the curved metal surface quenches a perpendicular relative to a parallel dipole; it is about three for hemispherical tips and the equal-volume sphere, decreases towards the flat-surface value of two for flatter ends and grows for sharper tips, but does not depend on the plasmon resonance. On top of it, a single mode—the longitudinal dipolar plasmon—contributes a resonant factor that grows with the oscillator strength of that mode, from 19 to 872 between aspect ratios 2 and 5, and is insensitive to the tip shape. The equal-volume control and the eigenmode decomposition lead independently to this separation, and the decomposition explains why the anisotropy peaks at an intermediate gap: the background is shared by both orientations and dominates at small gaps, whereas the coupling to the mode decays more slowly.

For randomly oriented emitters on the axis, $\langle F_p \rangle = (F_p^{\parallel} + 2F_p^{\perp})/3 = 555$ and $\langle T \rangle = 39$ at 608 nm and a 5 nm gap, three times below the axial values, while the averaged efficiency (7.0 %) is close to the axial one. The ensemble is bimodal: the transverse two thirds of the orientations are quenched ($\eta_a = 1.1$ %, $T = 0.35$). An enhancement factor measured on an orientation ensemble [28, 29] therefore hides a population of nearly dark emitters, which matters when each emitter’s orientation is fixed.

For single-photon sources the results give simple rules: a gap of 5–10 nm for poor emitters, for which the antenna raises the efficiency, and 15 nm or more for near-unity emitters, for which it mainly accelerates the emission; an emission wavelength well to the red of the interband edge (above about 650 nm for gold), which favours longer and thicker rods, flatter ends and a glass substrate; resonant excitation if an excitation enhancement is wanted; and a rod lying in the focal plane of the collection optics. Methodologically, volume-discretisation solvers reproduce the radiated power, the resonance and the emission pattern, but statements about quenching and the optimal gap require a surface method or careful convergence studies.

The limitations of the present study are the local-response description of gold beyond the hydrodynamic estimate, which is adequate at gaps of a few nanometres but not below about 1 nm [23, 55, 56]; surface scattering [57, 58]; the point-dipole, weak-coupling description of the emitter, which Table 3 justifies; and the quasistatic eigenmode analysis, which shifts the resonance by 13 nm and gives anisotropy ratios about 35 % lower than the retarded ones while reproducing

their gap dependence. The retarded pole fits confirm the dominance of the dipolar mode but do not resolve the background; a full quasinormal-mode expansion [45, 46] would remove this limitation.

6 Conclusion

Using converged boundary-element calculations, an equal-volume sphere as a control and a plasmon-eigenmode decomposition, we have separated the orientational anisotropy of the Purcell enhancement near a gold nanorod tip into a geometric factor set by the curvature of the metal surface (about three for hemispherical tips and the equal-volume sphere) and a resonant factor carried by the longitudinal dipolar mode, which reaches 49 in the total and 329 in the radiative rate for an aspect-ratio-3 rod at a 5 nm gap, peaks near a 10 nm gap and grows with the aspect ratio. More than 92 % of the emitted power is absorbed at the Purcell maximum for a 5 nm gap; the optimal gap for a single-photon source depends on the intrinsic quantum yield of the emitter, and resonance-matched rods emitting above 650 nm, on glass and in the focal plane of the collection optics, offer the best compromise between speed, brightness and efficiency. The model reproduces, without adjustable parameters, the measured thousandfold enhancement of single-molecule fluorescence near gold nanorods. Simulation scripts, raw spectra and analysis code are openly available.

Data availability

All raw spectra, simulation scripts (MNPBEM, Lumerical and Meep) and analysis code are available at <https://github.com/aminsadidi/maghale> (a Zenodo DOI will be assigned at publication).

Author contributions

[to be completed by the authors.]

Conflict of interest

The authors declare no competing interests.

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